

- (i) State an expression, in terms of some or all of m , Q , B and v , for the magnetic force F that acts on the particle when it is at point Y.

$$F = \dots\dots\dots [1]$$

- (ii) On Fig. 7.1, draw an arrow at point Y to indicate the direction of the force in **(b)(i)**. [1]
- (iii) On Fig. 7.1, draw a line to show a possible path for the particle through the region of the magnetic field. [1]
- (c) (i) Explain how an electric field can be used with the magnetic field to ensure that the particle in **(b)** now passes through point Z.

.....

.....

.....

.....

..... [3]

- (ii) Derive an expression for v in terms of B and the electric field strength E .

$$v = \dots\dots\dots [2]$$

[Total: 10]

